

FACTORY FLOOR OPTIMIZER

TECHNICAL DOCUMENTATION & USER MANUAL

Project Manager	Joey (System Orchestration & ReportLab Reporting)
Python Architect	Samantha (Web GUI, HTTP Server, Ctypes Bridge)
C Systems Engineer	Robert (High-Performance Simulation & Geometry Engine)
Documentation Date	October 2024 / October 2026 Revision
Target Audience	Manufacturing Facility Engineers & Layout Designers

Executive Summary:

This document serves as the complete engineering reference, software architecture overview, and operational manual for the Factory Floor Optimizer software.

1. System Architecture & High-Level Design

The software package is organized into a modular structure that divides high-performance geometric math from web layout rendering.

Module / File	Developer	Responsibility	Description
optimizer.h	Robert	Data Structures	Defines structs for Point, MachineInstance, FlowPath, CraneZones, etc.
optimizer.c	Robert	Geometry Math	Contains functions to check overlapping bounding boxes, line intersections, etc.
c_backend_bridge.py	Samantha	Ctypes Middleware	Uses Python ctypes to marshal Python dicts into C memory-aligned structs.
app.py	Samantha	Server & API	Runs standard Python HTTP service, serves JSON endpoints, handles requests.
index.html	Samantha	Interactive Web GUI	Provides HTML5 canvas, sidebar, live asset creation, and standard web controls.
run.py	Joey	Deployment script	Checks compile tools, compiles backend with a Makefile, tests and runs the application.